

Question Number	Answer	Additional guidance	Mark
7(a)	An explanation that makes reference to three of the following points: • (the body cells of) baby will have {26 / half the} chromosomes that are {similar / identical} to those in the mothers (cells) (1) • the other {26 / half of} chromosomes in the baby body cells will be (genetically) different as they came from the {sperm cell / father} (1) • (there may be genetic differences in the gamete chromosomes inherited from the mother) due to {mutation / crossing over} (1)	ACCEPT {genetic material / DNA / alleles} for chromosomes IGNORE genes ACCEPT baby has inherited {26 / half} of the chromosomes from the mother IGNORE genetic similarities unqualified IGNORE inherited some chromosomes from mother ecf for 13 ACCEPT baby has inherited {26 / half} of the chromosomes from the father IGNORE inherited some chromosomes from father IGNORE {independent / random} assortment IGNORE incorrect stage of meiosis	(3)

Question Number	Answer	Mark
7(b)(i)	The only correct answer is A number of heterozygotes ÷ number of individuals in the population <i>B is not correct because the equation is number of heterozygotes ÷ number of individuals in the population</i> <i>C is not correct because the equation is number of heterozygotes ÷ number of individuals in the population</i> <i>D is not correct because the equation is number of heterozygotes ÷ number of individuals in the population</i>	(1)

Question Number	Answer	Additional guidance	Mark
7(b)(ii)	An explanation that makes reference to two of the following points: - count the number of tapirs and {melanistic / all black} tapirs in the population (1) - use the Hardy-Weinberg equation to calculate the recessive allele frequency (1) - compare to previous recessive allele frequency to see if it has changed (1)	ACCEPT determine the number of {melanistic / all black} tapirs and the population size ACCEPT correct Hardy-Weinberg equation ACCEPT calculate {for each generation / at two(+) different time periods} and see if there is a change (over time) IGNORE see if there is a change in frequency IGNORE q^2	(2)

Question Number	Answer	Additional guidance	Mark
7(c)(i)	still found in {the same countries / area / region / islands / Thailand / Sumatra / Malaysia} (1)	ACCEPT still found in {central / highland} regions of the islands ACCEPT still found within earlier distribution area ACCEPT both still not found in {Cambodia / Vietnam}	(1)

Question Number	Answer	Additional guidance	Mark
7(c)(ii)	An answer that makes reference to the following point: reason as to why human activity changed distribution (1)	E.g (deforestation resulting in) {isolated pockets of forest / habitat fragmentation / geographical isolation / loss of habitat / loss of food / disruption to food chain / migration / separation of populations e.g. reduction in the numbers of {tapirs / territories} (due to hunting) / tapir moved to other areas (due to hunting) IGNORE answers referring to extinction of tapir IGNORE hunting / deforestation / reduction in biodiversity	(1)

Question Number	Answer	Additional guidance	Mark
7(c)(iii)	An explanation that makes reference to three of the following points: • analyse {individual's alleles / gene pool} / identify individuals with different alleles (1) • use a stud book (1) • breed {individuals / tapirs} {with different alleles / from different populations} (1) • (therefore) preventing loss of alleles (from gene pool) (1)	IGNORE genes ACCEPT identify individuals which are (more) genetically different ACCEPT DNA analysis / molecular phylogeny IGNORE genes / gene pools ACCEPT transfer {males / females / tapirs / sperm} from one {isolated population / zoo} to another ACCEPT breed individuals that are not closely related ACCEPT promote outbreeding / prevent inbreeding IGNORE prevent hybrids / interbreeding	(3)